

1. Administrative & Study Identification

Full Study Title: A Phase II, Open-label, Single-arm Study of Osimertinib as Induction Therapy Prior to CRT and Maintenance Osimertinib in Patients With Epidermal Growth Factor Receptor (EGFR) Mutation-positive, Stage III, Unresectable Non-small Cell Lung Cancer (NEOLA) **Brief Title / Acronym:** To Evaluate the Efficacy/Safety of Osimertinib Prior to CRT and Maintenance of it With Stage III, Unresectable NSCLC With EGFR Mutations (NEOLA) **Protocol Number:** D516AC00003 **NCT Number:** NCT06194448 **Trial Status:** Active, Not Recruiting **Sponsor Name:** AstraZeneca **Investigational Product:** Osimertinib (combined with Chemoradiotherapy [CRT]: Cisplatin/Carboplatin + Pemetrexed/Paclitaxel + Radiation) **Phase of Development:** Phase II **Disease Area:** Lung Cancer (Non-Small Cell Lung Carcinoma) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To measure the efficacy and safety of osimertinib as an induction therapy prior to curative intent CRT and maintenance osimertinib in adult patients with Stage III, unresectable NSCLC harboring common EGFR mutations (exon 19 deletion or L858R, including de novo T790M). **Secondary Objectives:** To evaluate Objective Response Rate (ORR), Disease Control Rate (DCR), Overall Survival (OS), and the overall safety and tolerability profile.

2.2 Background & Rationale Approximately 20-30% of patients with NSCLC present with locally advanced, stage III disease at the time of diagnosis, and 60-90% of these cases are unresectable. Osimertinib is a third-generation, CNS-active EGFR-TKI with proven efficacy. The NEOLA study builds upon prior data to generate clinical evidence evaluating if introducing osimertinib as an induction treatment prior to definitive chemoradiotherapy (CRT) improves outcomes for this specific patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 76 patients **Number of Sites:** 71 sites across 10 countries **Study Duration:** Approximately 2 years for recruitment and 2 years of

follow-up from the last patient's initiation. **Estimated Primary Completion:** 2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients must be ≥ 18 years or the legal age of consent in their jurisdiction at the time of signing the ICF. 2. Histologically documented NSCLC of predominantly non-squamous, squamous, or adenosquamous pathology presenting with locally advanced, unresectable (Stage III) disease. 3. Documented common EGFR mutations (exon 19 deletion or L858R, alone or in combination with other mutations including de novo T790M). 4. Eligible for and planning to undergo CCRT or SCRT treatment. 5. WHO performance status of 0 or 1 with no deterioration over the previous 2 weeks. 6. Minimum life expectancy of > 12 weeks at Day 1. 7. At least one lesion accurately measurable at baseline as ≥ 10 mm in the longest diameter (lymph nodes ≥ 15 mm short axis) via CT or MRI. 8. Patients who recurred from Stage I/II/III after complete surgery or gross incomplete resections are permitted if they have not received prior systemic/radiation therapy. 9. Patients with adequately controlled HBV or HIV co-infections are eligible under specific virologic and immunological protocol criteria. 10. Provision of signed Optional Genomics Initiative Research consent (declining does not exclude the patient from the main study).

4.2 Exclusion Criteria 1. Any presence of small cell and mixed small-cell and non-small cell histology. 2. Past medical history of interstitial lung disease (ILD)/pneumonitis, drug-induced ILD, radiation pneumonitis that required steroid treatment, or evidence of clinically active ILD/pneumonitis. 3. Unresolved toxicities from prior therapy greater than CTCAE Grade 1 (with exceptions for alopecia and prior Grade 2 platinum-therapy related neuropathy). 4. Patient meets strict cardiac criteria, including mean resting QTc > 470 msec, clinically important ECG abnormalities, or factors increasing the risk of QTc prolongation/TdP. 5. Inadequate bone marrow reserve or organ function (e.g., Absolute Neutrophil Count $< 1.5 \times 10^9/L$, Platelet count $< 100 \times 10^9/L$, Haemoglobin < 90 g/L, ALT/AST > 2.5 times ULN, Total bilirubin > 1.5 times ULN, or Creatinine > 1.5 times ULN). 6. Prior exposure to EGFR-TKI therapy or any prior treatment (chemotherapy, RT, immunotherapy, investigational) for locally advanced, unresectable Stage III NSCLC. 7. Concurrent use or inability to stop medications/herbal supplements known to be strong inducers of CYP3A4. 8. History of another primary malignancy ≤ 2 years before the first dose (with standard exceptions for curatively treated low-risk cancers). 9. Pregnant or breastfeeding females. 10. Major surgical procedure within 4 weeks of the first dose.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: - *Induction:* Osimertinib 80 mg once daily (QD) for up to 8 weeks. - *CRT Phase:* Platinum-based Chemotherapy (e.g., cisplatin) + Radiation for 6 weeks. - *Maintenance:* Osimertinib 80 mg QD until disease progression (PD) or unacceptable toxicity. **Route of Administration:** Oral (Osimertinib) and Intravenous/Targeted Radiation (CRT). **Duration of Treatment:** 8-week induction, 6-week CRT phase, followed by continuous maintenance until progression or death.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing CRT side effects. **Prohibited:** Medications known to prolong the QT interval and cause Torsades de Pointes (TdP), and strong inducers of CYP3A4.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening	Baseline Informed Consent, Medical History, Tumor Scans, EGFRm confirmation
2-9	Induction	Weeks 1-8 Osimertinib 80mg QD. Visits every 2 to 4 weeks. Scans at Week 8.
10-15	CRT Phase	Weeks 9-14 Chemoradiotherapy administration. Visits every 3 weeks for chemo; daily for RT.
16-24	Maintenance	Post-CRT Osimertinib Maintenance. Visits every 12 weeks until disease progression.

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from the date of the first dose until progression per RECIST 1.1 as assessed by the investigator at the local site, or death due to any cause.

7.2 Secondary & Exploratory Endpoints - Objective Response Rate (ORR) - Disease Control Rate (DCR) - Overall Survival (OS) - Treatment-emergent adverse events (safety and tolerability)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection during regular protocol visits (every 2-4 weeks during induction, every 3 weeks/daily during CRT, and every 12 weeks in maintenance). **Serious Adverse Events (SAEs):** Must be reported within standard 24-hour timelines following site awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by the sponsor (AstraZeneca).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT)/Full Analysis Set for efficacy endpoints. Safety Analysis Set for safety parameters.

Sample Size Justification: Target enrollment of 76 patients, sized to provide adequate power ($\alpha = 0.05$) to measure meaningful changes in PFS and ORR compared to historical CRT benchmarks. **Handling of Missing Data:** Standard survival analysis methods including censoring for PFS and OS at the time of last evaluable disease assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standardized clinical monitoring during induction, active CRT, and quarterly maintenance phases. **Audit Trail:** eCRF locking, tracking of protocol deviations, and source data verification managed via standard auditing mechanisms.

11. Study Identifiers & Governance

Primary ID: D516AC00003 **Registry Number:** NCT06194448 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** NEOLA **Official Regulatory Title:** A Phase II, Open-label, Single-arm Study of Osimertinib as Induction Therapy Prior to CRT and Maintenance Osimertinib in Patients With Epidermal Growth Factor Receptor (EGFR) Mutation-positive, Stage III, Unresectable Non-small Cell Lung Cancer (NEOLA).

12. Clinical Framework (Oncology/NSCLC Specific)

Primary Condition: Non-small Cell Lung Cancer (Stage III, Unresectable) **Diagnosis Threshold:** Histologically documented NSCLC harboring common EGFR mutations (exon 19 deletion or L858R) **Medical Methodology:** Induction TKI Therapy $\rightarrow$ Curative intent Chemoradiotherapy (CRT) $\rightarrow$ TKI Maintenance **Optimization Target:** Maximal tumor downstaging prior to CRT and prolonged PFS post-CRT

13. Study Procedures & Methodology

Management Pathway: Standard oncological RECIST 1.1 tumor assessment pathway **Education Protocol (HCP):** Site-initiation training on CRT integration and AE management. **Education Protocol (Patient):** Trial onboarding, oral TKI adherence education, and CRT toxicity self-monitoring. **Quality Audit Mechanism:** Local site

investigator radiologic review and standard sponsor-led data monitoring.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up continues until RECIST 1.1-defined progression or death. **Onsite Visit Cadence:** Every 2-4 weeks (Induction), every 3 weeks/daily (CRT), every 12 weeks (Maintenance). **Remote Monitoring Frequency:** Continuous patient reporting as required. **Checklist Review Interval:** Scans every 8 to 12 weeks for PFS assessment.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				